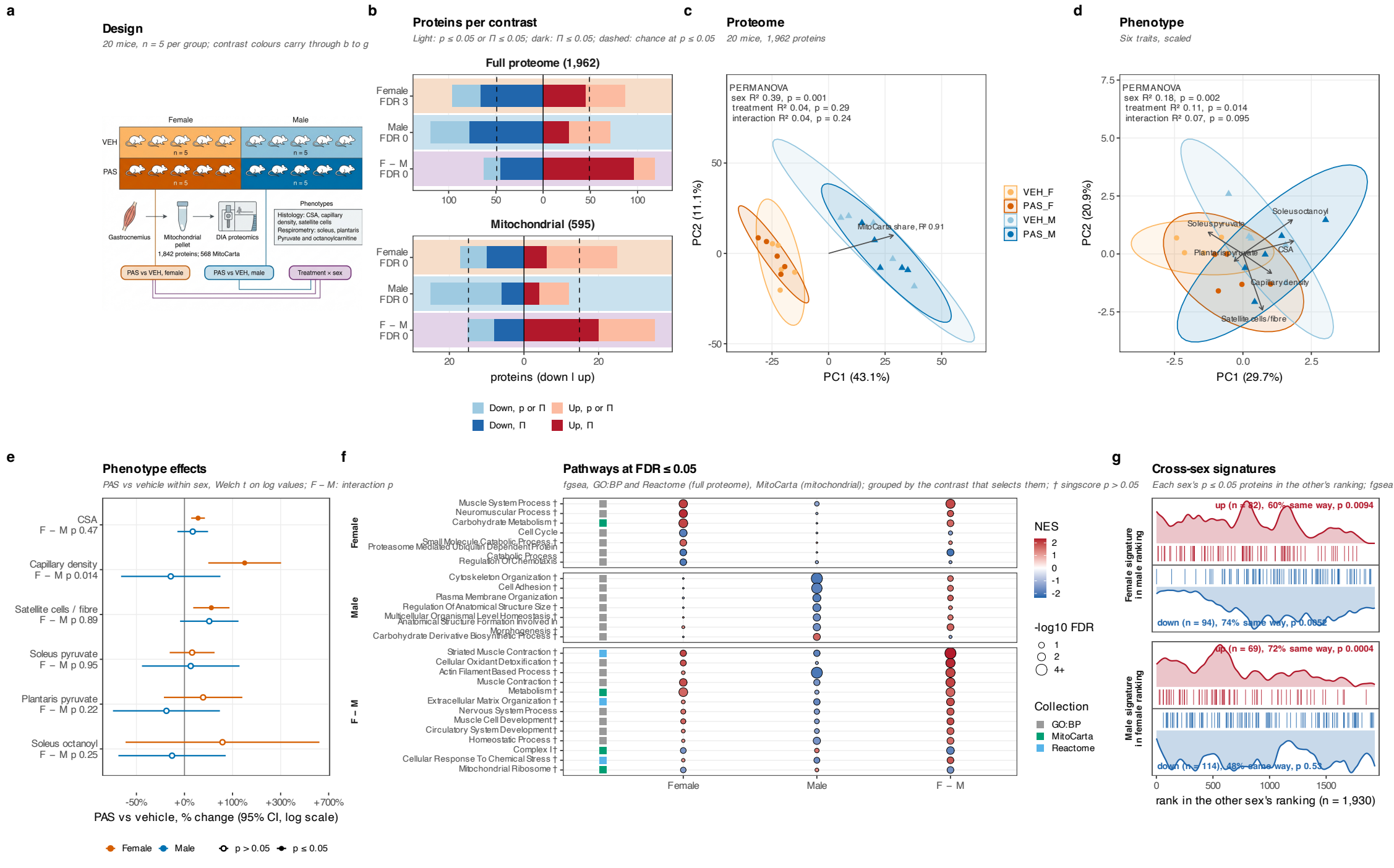

a
Design
20 mice, n = 5 per group; contrast colours carry through b to g
Female
Male
VEH
PAS
n = 5
Gastrocnemius
Mitochondrial pellet
DIA proteomics
1,842 proteins; 568 MitoCarta
Phenotypes
Histology: CSA, capillary density, satellite cells
Respirometry: soleus, plantaris
Pyruvate and octanoylcarnitine
PAS vs VEH, female
PAS vs VEH, male
Treatment × sex
b
Proteins per contrast
Light: p ≤ 0.05 or Π ≤ 0.05; dark: Π ≤ 0.05; dashed: chance at p ≤ 0.05
Full proteome (1,962)
Female FDR 3
Male FDR 0
F – M FDR 0
Mitochondrial (595)
Female FDR 0
Male FDR 0
F – M FDR 0
proteins (down | up)
Down, p or Π
Up, p or Π
Down, Π
Up, Π
c
Proteome
20 mice, 1,962 proteins
PERMANOVA
sex R² 0.39, p = 0.001
treatment R² 0.04, p = 0.29
interaction R² 0.04, p = 0.24
MitoCarta share, R² 0.91
PC1 (43.1%)
PC2 (11.1%)
VEH_F
PAS_F
VEH_M
PAS_M
d
Phenotype
Six traits, scaled
PERMANOVA
sex R² 0.18, p = 0.002
treatment R² 0.11, p = 0.014
interaction R² 0.07, p = 0.095
Soleus octanoyl
Soleus pyruvate
Plantaris pyruvate
CSA
Capillary density
Satellite cells / fibre
PC1 (29.7%)
PC2 (20.9%)
e
Phenotype effects
PAS vs vehicle within sex, Welch t on log values; F – M: interaction p
CSA
F – M p 0.47
Capillary density
F – M p 0.014
Satellite cells / fibre
F – M p 0.89
Soleus pyruvate
F – M p 0.95
Plantaris pyruvate
F – M p 0.22
Soleus octanoyl
F – M p 0.25
-50%
+0%
+100%
+300%
+700%
PAS vs vehicle, % change (95% CI, log scale)
Female
Male
p > 0.05
p ≤ 0.05
f
Pathways at FDR ≤ 0.05
fgsea, GO:BP and Reactome (full proteome), MitoCarta (mitochondrial); grouped by the contrast that selects them; † singscore p > 0.05
Female
Male
F – M
Muscle System Process †
Neuromuscular Process †
Carbohydrate Metabolism †
Cell Cycle
Small Molecule Catabolic Process †
Proteasome Mediated Ubiquitin Dependent Protein Catabolic Process
Regulation Of Chemotaxis
Cytoskeleton Organization †
Cell Adhesion †
Plasma Membrane Organization
Regulation Of Anatomical Structure Size †
Multicellular Organismal Level Homeostasis †
Anatomical Structure Formation Involved In Morphogenesis †
Carbohydrate Derivative Biosynthetic Process †
Striated Muscle Contraction †
Cellular Oxidant Detoxification †
Actin Filament Based Process †
Muscle Contraction †
Metabolism †
Extracellular Matrix Organization †
Nervous System Process
Muscle Cell Development †
Circulatory System Development †
Homeostatic Process †
Complex I †
Cellular Response To Chemical Stress †
Mitochondrial Ribosome †
NES
-log10 FDR
Collection
GO:BP
MitoCarta
Reactome
g
Cross-sex signatures
Each sex's p ≤ 0.05 proteins in the other's ranking; fgsea
Female signature in male ranking
up (n = 82), 60% same way, p 0.0094
down (n = 94), 74% same way, p 0.0052
Male signature in female ranking
up (n = 69), 72% same way, p 0.0004
down (n = 114), 48% same way, p 0.53
rank in the other sex's ranking (n = 1,930)